there are ways of escaping from the circularity, but this book is not the place to pursue the question. Sex exists. That much is true. It is a consequence of sex and crossing-over that the small genetic unit or gene can be regarded as the nearest thing we have to a fundamental, independent agent of evolution.

Sex is not the only apparent paradox that becomes less puzzling the moment we learn to think in selfish gene terms. For instance, it appears that the amount of DNA in organisms is more than is strictly necessary for building them: a large fraction of the DNA is never translated into protein. From the point of view of the individual organism this seems paradoxical. If the 'purpose' of DNA is to supervise the building of bodies, it is surprising to find a large quantity of DNA which does no such thing. Biologists are racking their brains trying to think what useful task this apparently surplus DNA is doing. But from the point of view of the selfish genes themselves, there is no paradox. The true 'purpose' of DNA is to survive, no more and no less. The simplest way to explain the surplus DNA is to suppose that it is a parasite, or at best a harmless but useless passenger, hitching a ride in the survival machines created by the other DNA.*

Some people object to what they see as an excessively genecentred view of evolution. After all, they argue, it is whole individuals with all their genes who actually live or die. I hope I have said enough in this chapter to show that there is really no disagreement here. Just as whole boats win or lose races, it is indeed individuals who live or die, and the *immediate* manifestation of natural selection is nearly always at the individual level. But the long-term consequences of non-random individual death and reproductive success are manifested in the form of changing gene frequencies in the gene pool. With reservations, the gene pool plays the same role for the modern replicators as the primeval soup did for the original ones. Sex and chromosomal crossing-over have the effect of preserving the liquidity of the modern equivalent of the soup. Because of sex and crossing-over the gene pool is kept well stirred, and the genes partially shuffled. Evolution is the process by which some genes become more numerous and others less numerous in the gene pool. It is good to get into the habit, whenever we are trying to explain the evolution of some characteristic, such as altruistic behaviour, of asking ourselves simply: What effect will this characteristic have on frequencies of genes in the gene pool? At times, gene language gets a bit tedious, and